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The (25, 12)-System

N = 25A + 12B | 25 = 1 (mod 12) | The Clock Structure

Een volledige analyse van het lineaire Diophantische representation system N = 25A + 12B, analogous to the (19,9)-system. The minimal A-coordinate always equals N mod 12. Exact O(1) counting formula. Frobenius boundary = 263. Verified for N = 1 to 1000.

Par.Par.0 -- What is a representation

A representation of N in the system (25, 12) is a pair of non-negative integers (A, B) such that 25A + 12B = N.

Denk aan biljetten van EUR25 en EUR12. Op hoeveel manieren kun je precies EUR575 betalen Het antwoord is 2 -- en je vindt ze allemaal in een berekening.

KEY INSIGHT -- WHY 25 = 1 (mod 12)

25 = 2 x 12 + 1, so 25 = 1 (mod 12). Take N = 25A + 12B modulo 12:

$$N \bmod 12 = (25A + 12B) \bmod 12 = (1|A + 0|B) \bmod 12 = A \bmod 12$$

B vanishes completely from the equation mod 12! Therefore A0 = N mod 12 -- directly determinable in one step.

Concept	Meaning	Example
mod (modulo)	Remainder after division	17 mod 12 = 5
p = 1 (mod q)	p divided by q leaves remainder 1	25 = 1 (mod 12)
A0	Minimal A-coordinate	A0 = N mod 12
R(N)	Number of representations of N	R(575) = 2
floor(N/25)	Round down to nearest integer	floor(575/25) = 23

Par.Par.1 -- The Main Theorem: A0 = N mod 12

A0 = N mod 12 voor alle N >= 300

De kleinste A waarvoor N = 25A + 12B een niet-negatieve oplossing heeft, is altijd gelijk aan N mod 12. Alle oplossingen vormen een regelmatige ladder:

$$A = A0 + 12|k,\ B = (N - 25|A) / 12 \text{ voor } k = 0, 1, ..., k_{max}$$

where A0 = N mod 12. Each step increases A by 12 and decreases B by 25.

PROOF -- 3 STEPS

Stap 1: Neem N = 25A + 12B modulo 12. Omdat 12B = 0 (mod 12) en 25 = 1 (mod 12): N = A (mod 12). Dus A moet congruent zijn aan N mod 12.

Stap 2: De kleinste niet-negatieve A met deze congruentie is A0 = N mod 12. Omdat N >= 300 >= 25|A0, is B = (N - 25A0)/12 >= 0.

Step 3: All other representations arise via A -> A+12, B -> B-25. This preserves 25A + 12B = N. We stop when B would become negative.

All 2 Representations of N = 575 in System (25, 12)

#	A	B	25xA	12xB	Som	A mod 12	Remark
1	11	25	275	300	575	11	A0 = 575 mod 12 = 11 <- start
2	23	0	575	0	575	11	Last: B becomes 0

Par.Par.2 -- De Gesloten Counting formula R(25, 12; N)

R(25, 12; N) = floor((floor(N/25) - A0) / 12) + 1

Deze formule is **exact** -- geen benadering. Twee bewerkingen: (1) bereken A0 = N mod 12, (2) pas de vloerformule toe. Dat is O(1).

Asymptotically: R(N) ~ N / (25 x 12) = N / 300 for large N.

N	Keten	floor(N/25)	A0 = N mod 12	Calculation	R(N)
300	300->3->3	12	0	floor((12-0)/12)+1	2
575	575->17->17	23	11	floor((23-11)/12)+1	2
600	600->6->6	24	0	floor((24-0)/12)+1	3
900	900->9->9	36	0	floor((36-0)/12)+1	4
1000	1000->10->1	40	4	floor((40-4)/12)+1	4
1200	1200->3->3	48	0	floor((48-0)/12)+1	5

Par.Par.3 -- Frobenius Grens en Familie-Grenzen

Frobenius boundary (25,12) = 263 -- N = 11 (mod 12), the last residue

After N = 263, every integer has a representation in the (25,12)-system. This is analogous to the boundary 143 in the (19,9)-system. Opmerkelijk: net als dr=8 de last digital root is die een representation krijgt in (19,9), is r=11 de last residue class in (25,12) -- N=275 is the first N=11 (mod 12) with a representation.

FAMILY BOUNDARIES G(k)

Voor een vaste residue r is de t-th boundary T(t,r) = 25r + 300(t-1). Elke keer dat het minimale aantal representations met 1 stijgt, verschuift the boundary met precies 300 = 25 x 12 = p x q.

$$G(k) = 300(k-1) + 12 = 300k - 288 \text{ (voor het (25,12)-systeem)}$$

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G(1) = 12 <- Frobenius+1: first representation G(2) = 300 <- first N with >=2
representations G(10) = 2712 <- from here every N has >=10 representations
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k	G(k)	Meaning
1	12	First representation (N=12 = 0x25 + 1x12)
2	300	First N with >=2 representations
3	600	First N with >=3 representations
4	900	
5	1200	
6	1500	
10	2712	From here: every N has >=10 representations

First representation per residue class:

r = N mod 12	First N	Via	Remark
0	12	0x25 + 1x12	Pure B
1	25	1x25 + 0x12	Pure A
2	50	2x25 + 0x12	
3	75	3x25 + 0x12	
4	100	4x25 + 0x12	
5	125	5x25 + 0x12	
6	150	6x25 + 0x12	
7	175	7x25 + 0x12	
8	200	8x25 + 0x12	
9	225	9x25 + 0x12	
r = N mod 12	First N	Via	Remark
10	250	10x25 + 0x12	
11	275	11x25 + 0x12	Last -- analogous to dr=8 in (19,9)

Par.Par.4 -- De Property van r = 11: De Laatste Residue

In het (19,9)-systeem heeft dr=8 een bijzondere eigenschap: het is de last digital root die een representation krijgt, en bij grotere getallen heeft dr=8 structureel een representation minder dan omliggende getallen. In het (25,12)-systeem speelt r=11 (d.w.z. N = 11 mod 12) precies dezelfde rol.

PROOF OF THE ROLE OF r=11

De minimale A voor elk N = 11 (mod 12) is altijd **A=11**. Dit is de grootste mogelijke A0-waarde in het systeem (0 t/m 11). Daardoor is er minder 'ruimte' voor extra representations: de ladder A0, A0+12, A0+24, ... start later en eindigt even vroeg.

Verified: for N=311 through N=539 (outside the exception points) R(N) < R(N+1) when N = 11 (mod 12) -- exactly as R(1682) = 9 while surrounding numbers have 10 representations.

Comparison r=11 versus neighbours (selection):

N	N mod 12	R(N-1)	R(N)	R(N+1)	Fewer
311	11	1	1	2	JA
323	11	1	1	2	JA
395	11	1	1	2	JA
431	11	1	1	2	JA
503	11	1	1	2	JA
551	11	2	1	2	JA
575	11	2	2	2	no (exception point)
599	11	2	2	3	JA

De uitzonderingen (zoals N=575) zijn structureel: ze komen voor bij N = 25x11 en N = 25x11+12 -- de exacte grenspunten waarop een nieuw representationniveau begint. Geen willekeur, maar geometrie.

Par.Par.5 -- De Families: Eerste 2 Reps en de Ankerfamilie met 10 Reps

In het (19,9)-systeem zijn de getallen N=1675-1683 de ankerfamilie: negen opeenvolgende getallen where elke digitale wortel 1-9 precies een keer voorkomt en elk lid 10 representations heeft -- op een uitzondering na.

In het (25,12)-systeem zijn er twee families te benoemen: de first sequence met at least 2 representations (N=564-575), en de ankerfamilie met 10 representations (N=2964-2975).

First complete family: N = 564-575 (each >=2 representations)

N	r=N mod 12	A0	R(N)	All representations
564	0	0	2	0x25+47x12 | 12x25+22x12
565	1	1	2	1x25+45x12 | 13x25+20x12
566	2	2	2	2x25+43x12 | 14x25+18x12
567	3	3	2	3x25+41x12 | 15x25+16x12
568	4	4	2	4x25+39x12 | 16x25+14x12

569	5	5	2	5x25+37x12 | 17x25+12x12
570	6	6	2	6x25+35x12 | 18x25+10x12
571	7	7	2	7x25+33x12 | 19x25+8x12
572	8	8	2	8x25+31x12 | 20x25+6x12
573	9	9	2	9x25+29x12 | 21x25+4x12
574	10	10	2	10x25+27x12 | 22x25+2x12
575	11	11	2	11x25+25x12 | 23x25+0x12

The Anchor Family: N = 2964-2975 (each exactly 10 representations)

De twaalf opeenvolgende getallen 2964-2975 vormen de ankerfamilie van het (25,12)-systeem: elke residue r=0 t/m 11 verschijnt precies een keer, en elk lid heeft exact 10 representations. From N=2964 onward, every integer at least 10 representations.

N	r=N mod 12	dr(N)	R(N)	Minimal representation
2964	0	3	10	0x25 + 247x12
2965	1	4	10	1x25 + 245x12
2966	2	5	10	2x25 + 243x12
2967	3	6	10	3x25 + 241x12
2968	4	7	10	4x25 + 239x12
2969	5	8	10	5x25 + 237x12
2970	6	9	10	6x25 + 235x12
2971	7	1	10	7x25 + 233x12
2972	8	2	10	8x25 + 231x12
2973	9	3	10	9x25 + 229x12
2974	10	4	10	10x25 + 227x12
N	r=N mod 12	dr(N)	R(N)	Minimal representation
2975	11	5	10	11x25 + 225x12

THE GAP: N = 2963 (r=11) -- 9 representations

N = 2963 heeft slechts **9 representations**, terwijl alle omringende getallen 10 hebben. Dit is het directe equivalent van N=1682 in het (19,9)-systeem (dr=8, ook 9 representations terwijl de rest 10 had).

Oorzaak: N=2963 heeft r=11 -- de last residue class. R(2963) = floor((floor(2963/25) - 11)/12) + 1 = floor((118-11)/12) + 1 = floor(107/12) + 1 = 8+1 = 9.

Dit is geen fout of anomalie -- het is structureel bepaald door r=11 als de last residue. Het gat sluit op N=2975, waar r=11 eindelijk zijn 10e representation bereikt en de familie compleet is.

N	r	R(N)	Remark
2961	9	10	
2962	10	10	
2963	11	9	THE GAP -- analogous to N=1682 in (19,9)

2964	0	10	Start of anchor family
2965	1	10	
...	...	...	
2974	10	10	
2975	11	10	End of anchor family -- r=11 closes the sequence
2976	0	10	From here: every N >=10 representations

Par.Par.6 -- Vergelijking: (19,9)-systeem versus (25,12)-systeem

Beide systemen delen dezelfde algebraische kern: p = 1 (mod q), waardoor B verdwijnt uit de vergelijking modulo q. De structuur is identiek -- alleen de taal verschilt: digitale wortel versus residue mod 12.

Property	(19,9)-systeem	(25,12)-systeem
Condition	19 = 1 (mod 9)	25 = 1 (mod 12)
Minimale A (A0)	A0 = N mod 9 = dr(N)	A0 = N mod 12
Counting formula	floor((floor(N/19) - dr(N))/9) + 1	floor((floor(N/25) - N mod 12)/12) + 1
Frobenius boundary	143	263
Last residue	dr=8 (digital root)	r=11 (N mod 12)
pxq (periode)	171	300
G(k) formule	G(k) = 171k - 27	G(k) = 300k - 288
Property	(19,9)-systeem	(25,12)-systeem
First complete family	N=1675-1683 (9 integers)	N=564-575 (12 integers)
Clock analogy	10 = 1 (mod 9)	13 = 1 (mod 12)
Symmetry points	dr(A)=dr(B)=dr(N)	A mod 12 = B mod 25 = N mod 12

THE CLOCK ANALOGY

Het meest alledaagse voorbeeld van modulaire rekenkunde is de klok: na 12 komt 1. Dit is precies 13 = 1 (mod 12). In het (25,12)-systeem is 25 = 1 (mod 12) de motor: after 25 steps the cycle restarts. The same structure als 10 = 1 (mod 9) with the digital root -- de reden dat adding digits preserves the remainder mod 9.

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10 = 1 (mod 9) -> digitale wortel -> (19,9)-systeem 13 = 1 (mod 12) -> de klok ->
(25,12)-systeem
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Par.Par.7 -- Representatietabel N = 1 tot 1000

Counting formula: R(N) = floor((floor(N/25) - N mod 12) / 12) + 1 als floor(N/25) >= N mod 12, anders R(N) = 0.

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
1	1	1	0	not representable
2	2	2	0	not representable
3	3	3	0	not representable
4	4	4	0	not representable
5	5	5	0	not representable
6	6	6	0	not representable
7	7	7	0	not representable
8	8	8	0	not representable
9	9	9	0	not representable
10	10	10	0	not representable
11	11	11	0	not representable
12	0	0	1	0x25+1x12
13	1	1	0	not representable
14	2	2	0	not representable
15	3	3	0	not representable
16	4	4	0	not representable
17	5	5	0	not representable
18	6	6	0	not representable
19	7	7	0	not representable
20	8	8	0	not representable
21	9	9	0	not representable
22	10	10	0	not representable
23	11	11	0	not representable
24	0	0	1	0x25+2x12
25	1	1	1	1x25+0x12
26	2	2	0	not representable
27	3	3	0	not representable
28	4	4	0	not representable
29	5	5	0	not representable
30	6	6	0	not representable
31	7	7	0	not representable
32	8	8	0	not representable
33	9	9	0	not representable
34	10	10	0	not representable
35	11	11	0	not representable
36	0	0	1	0x25+3x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
37	1	1	1	1x25+1x12
38	2	2	0	not representable
39	3	3	0	not representable
40	4	4	0	not representable
41	5	5	0	not representable
42	6	6	0	not representable
43	7	7	0	not representable
44	8	8	0	not representable
45	9	9	0	not representable
46	10	10	0	not representable
47	11	11	0	not representable
48	0	0	1	0x25+4x12
49	1	1	1	1x25+2x12
50	2	2	1	2x25+0x12
51	3	3	0	not representable
52	4	4	0	not representable
53	5	5	0	not representable
54	6	6	0	not representable
55	7	7	0	not representable
56	8	8	0	not representable
57	9	9	0	not representable
58	10	10	0	not representable
59	11	11	0	not representable
60	0	0	1	0x25+5x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
61	1	1	1	1x25+3x12
62	2	2	1	2x25+1x12
63	3	3	0	not representable
64	4	4	0	not representable
65	5	5	0	not representable
66	6	6	0	not representable
67	7	7	0	not representable

68	8	8	0	not representable
69	9	9	0	not representable
70	10	10	0	not representable
71	11	11	0	not representable
72	0	0	1	0x25+6x12
73	1	1	1	1x25+4x12
74	2	2	1	2x25+2x12
75	3	3	1	3x25+0x12
76	4	4	0	not representable
N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
77	5	5	0	not representable
78	6	6	0	not representable
79	7	7	0	not representable
80	8	8	0	not representable
81	9	9	0	not representable
82	10	10	0	not representable
83	11	11	0	not representable
84	0	0	1	0x25+7x12
85	1	1	1	1x25+5x12
86	2	2	1	2x25+3x12
87	3	3	1	3x25+1x12
88	4	4	0	not representable
89	5	5	0	not representable
90	6	6	0	not representable
91	7	7	0	not representable
92	8	8	0	not representable
93	9	9	0	not representable
94	10	10	0	not representable
95	11	11	0	not representable
96	0	0	1	0x25+8x12
97	1	1	1	1x25+6x12
98	2	2	1	2x25+4x12
99	3	3	1	3x25+2x12
100	4	4	1	4x25+0x12
101	5	5	0	not representable
102	6	6	0	not representable
103	7	7	0	not representable
104	8	8	0	not representable

105	9	9	0	not representable
106	10	10	0	not representable
107	11	11	0	not representable
108	0	0	1	0x25+9x12
109	1	1	1	1x25+7x12
110	2	2	1	2x25+5x12
111	3	3	1	3x25+3x12
112	4	4	1	4x25+1x12
113	5	5	0	not representable
114	6	6	0	not representable
115	7	7	0	not representable
116	8	8	0	not representable
117	9	9	0	not representable

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
118	10	10	0	not representable
119	11	11	0	not representable
120	0	0	1	0x25+10x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
121	1	1	1	1x25+8x12
122	2	2	1	2x25+6x12
123	3	3	1	3x25+4x12
124	4	4	1	4x25+2x12
125	5	5	1	5x25+0x12
126	6	6	0	not representable
127	7	7	0	not representable
128	8	8	0	not representable
129	9	9	0	not representable
130	10	10	0	not representable
131	11	11	0	not representable
132	0	0	1	0x25+11x12
133	1	1	1	1x25+9x12
134	2	2	1	2x25+7x12
135	3	3	1	3x25+5x12
136	4	4	1	4x25+3x12
137	5	5	1	5x25+1x12

138	6	6	0	not representable
139	7	7	0	not representable
140	8	8	0	not representable
141	9	9	0	not representable
142	10	10	0	not representable
143	11	11	0	not representable
144	0	0	1	0x25+12x12
145	1	1	1	1x25+10x12
146	2	2	1	2x25+8x12
147	3	3	1	3x25+6x12
148	4	4	1	4x25+4x12
149	5	5	1	5x25+2x12
150	6	6	1	6x25+0x12
151	7	7	0	not representable
152	8	8	0	not representable
153	9	9	0	not representable
154	10	10	0	not representable
155	11	11	0	not representable
156	0	0	1	0x25+13x12
157	1	1	1	1x25+11x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
158	2	2	1	2x25+9x12
159	3	3	1	3x25+7x12
160	4	4	1	4x25+5x12
161	5	5	1	5x25+3x12
162	6	6	1	6x25+1x12
163	7	7	0	not representable
164	8	8	0	not representable
165	9	9	0	not representable
166	10	10	0	not representable
167	11	11	0	not representable
168	0	0	1	0x25+14x12
169	1	1	1	1x25+12x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
170	2	2	1	2x25+10x12
171	3	3	1	3x25+8x12
172	4	4	1	4x25+6x12
173	5	5	1	5x25+4x12
174	6	6	1	6x25+2x12
175	7	7	1	7x25+0x12
176	8	8	0	not representable
177	9	9	0	not representable
178	10	10	0	not representable
179	11	11	0	not representable
180	0	0	1	0x25+15x12
181	1	1	1	1x25+13x12
182	2	2	1	2x25+11x12
183	3	3	1	3x25+9x12
184	4	4	1	4x25+7x12
185	5	5	1	5x25+5x12
186	6	6	1	6x25+3x12
187	7	7	1	7x25+1x12
188	8	8	0	not representable
189	9	9	0	not representable
190	10	10	0	not representable
191	11	11	0	not representable
192	0	0	1	0x25+16x12
193	1	1	1	1x25+14x12
194	2	2	1	2x25+12x12
195	3	3	1	3x25+10x12
196	4	4	1	4x25+8x12
197	5	5	1	5x25+6x12
198	6	6	1	6x25+4x12
199	7	7	1	7x25+2x12
200	8	8	1	8x25+0x12
201	9	9	0	not representable

202	10	10	0	not representable
203	11	11	0	not representable
204	0	0	1	0x25+17x12
205	1	1	1	1x25+15x12
206	2	2	1	2x25+13x12
207	3	3	1	3x25+11x12
208	4	4	1	4x25+9x12
209	5	5	1	5x25+7x12
210	6	6	1	6x25+5x12
211	7	7	1	7x25+3x12
212	8	8	1	8x25+1x12
213	9	9	0	not representable
214	10	10	0	not representable
215	11	11	0	not representable
216	0	0	1	0x25+18x12
217	1	1	1	1x25+16x12
218	2	2	1	2x25+14x12
219	3	3	1	3x25+12x12
220	4	4	1	4x25+10x12
221	5	5	1	5x25+8x12
222	6	6	1	6x25+6x12
223	7	7	1	7x25+4x12
224	8	8	1	8x25+2x12
225	9	9	1	9x25+0x12
226	10	10	0	not representable
227	11	11	0	not representable
228	0	0	1	0x25+19x12
229	1	1	1	1x25+17x12
230	2	2	1	2x25+15x12
231	3	3	1	3x25+13x12
232	4	4	1	4x25+11x12
233	5	5	1	5x25+9x12
234	6	6	1	6x25+7x12
235	7	7	1	7x25+5x12
236	8	8	1	8x25+3x12
237	9	9	1	9x25+1x12
238	10	10	0	not representable

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
239	11	11	0	not representable
240	0	0	1	0x25+20x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
241	1	1	1	1x25+18x12
242	2	2	1	2x25+16x12
243	3	3	1	3x25+14x12
244	4	4	1	4x25+12x12
245	5	5	1	5x25+10x12
246	6	6	1	6x25+8x12
247	7	7	1	7x25+6x12
248	8	8	1	8x25+4x12
249	9	9	1	9x25+2x12
250	10	10	1	10x25+0x12
251	11	11	0	not representable
252	0	0	1	0x25+21x12
253	1	1	1	1x25+19x12
254	2	2	1	2x25+17x12
255	3	3	1	3x25+15x12
256	4	4	1	4x25+13x12
257	5	5	1	5x25+11x12
258	6	6	1	6x25+9x12
259	7	7	1	7x25+7x12
260	8	8	1	8x25+5x12
261	9	9	1	9x25+3x12
262	10	10	1	10x25+1x12
263	11	11	0	not representable
264	0	0	1	0x25+22x12
265	1	1	1	1x25+20x12
266	2	2	1	2x25+18x12
267	3	3	1	3x25+16x12
268	4	4	1	4x25+14x12
269	5	5	1	5x25+12x12
270	6	6	1	6x25+10x12
271	7	7	1	7x25+8x12
272	8	8	1	8x25+6x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
273	9	9	1	9x25+4x12
274	10	10	1	10x25+2x12
275	11	11	1	11x25+0x12
276	0	0	1	0x25+23x12
277	1	1	1	1x25+21x12
278	2	2	1	2x25+19x12
279	3	3	1	3x25+17x12
280	4	4	1	4x25+15x12
281	5	5	1	5x25+13x12
282	6	6	1	6x25+11x12
283	7	7	1	7x25+9x12
284	8	8	1	8x25+7x12
285	9	9	1	9x25+5x12
286	10	10	1	10x25+3x12
287	11	11	1	11x25+1x12
288	0	0	1	0x25+24x12
289	1	1	1	1x25+22x12
290	2	2	1	2x25+20x12
291	3	3	1	3x25+18x12
292	4	4	1	4x25+16x12
293	5	5	1	5x25+14x12
294	6	6	1	6x25+12x12
295	7	7	1	7x25+10x12
296	8	8	1	8x25+8x12
297	9	9	1	9x25+6x12
298	10	10	1	10x25+4x12
299	11	11	1	11x25+2x12
300	0	0	2	0x25+25x12 | 12x25+0x12
301	1	1	1	1x25+23x12
302	2	2	1	2x25+21x12
303	3	3	1	3x25+19x12

304	4	4	1	4x25+17x12
305	5	5	1	5x25+15x12
306	6	6	1	6x25+13x12
307	7	7	1	7x25+11x12
308	8	8	1	8x25+9x12
309	9	9	1	9x25+7x12
310	10	10	1	10x25+5x12
311	11	11	1	11x25+3x12
312	0	0	2	0x25+26x12 | 12x25+1x12
313	1	1	1	1x25+24x12
314	2	2	1	2x25+22x12
315	3	3	1	3x25+20x12
316	4	4	1	4x25+18x12
317	5	5	1	5x25+16x12
318	6	6	1	6x25+14x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
319	7	7	1	7x25+12x12
320	8	8	1	8x25+10x12
321	9	9	1	9x25+8x12
322	10	10	1	10x25+6x12
323	11	11	1	11x25+4x12
324	0	0	2	0x25+27x12 | 12x25+2x12
325	1	1	2	1x25+25x12 | 13x25+0x12
326	2	2	1	2x25+23x12
327	3	3	1	3x25+21x12
328	4	4	1	4x25+19x12
329	5	5	1	5x25+17x12
330	6	6	1	6x25+15x12
331	7	7	1	7x25+13x12
332	8	8	1	8x25+11x12
333	9	9	1	9x25+9x12
334	10	10	1	10x25+7x12
335	11	11	1	11x25+5x12
336	0	0	2	0x25+28x12 | 12x25+3x12
337	1	1	2	1x25+26x12 | 13x25+1x12
338	2	2	1	2x25+24x12
339	3	3	1	3x25+22x12

340	4	4	1	4x25+20x12
341	5	5	1	5x25+18x12
342	6	6	1	6x25+16x12
343	7	7	1	7x25+14x12
344	8	8	1	8x25+12x12
345	9	9	1	9x25+10x12
346	10	10	1	10x25+8x12
347	11	11	1	11x25+6x12
348	0	0	2	0x25+29x12 | 12x25+4x12
349	1	1	2	1x25+27x12 | 13x25+2x12
350	2	2	2	2x25+25x12 | 14x25+0x12
351	3	3	1	3x25+23x12
352	4	4	1	4x25+21x12
353	5	5	1	5x25+19x12
354	6	6	1	6x25+17x12
355	7	7	1	7x25+15x12
356	8	8	1	8x25+13x12
357	9	9	1	9x25+11x12
358	10	10	1	10x25+9x12
359	11	11	1	11x25+7x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
360	0	0	2	0x25+30x12 | 12x25+5x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
361	1	1	2	1x25+28x12 | 13x25+3x12
362	2	2	2	2x25+26x12 | 14x25+1x12
363	3	3	1	3x25+24x12
364	4	4	1	4x25+22x12
365	5	5	1	5x25+20x12
366	6	6	1	6x25+18x12
367	7	7	1	7x25+16x12
368	8	8	1	8x25+14x12
369	9	9	1	9x25+12x12
370	10	10	1	10x25+10x12
371	11	11	1	11x25+8x12
372	0	0	2	0x25+31x12 | 12x25+6x12
373	1	1	2	1x25+29x12 | 13x25+4x12

374	2	2	2	2x25+27x12 | 14x25+2x12
375	3	3	2	3x25+25x12 | 15x25+0x12
376	4	4	1	4x25+23x12
377	5	5	1	5x25+21x12
378	6	6	1	6x25+19x12
379	7	7	1	7x25+17x12
380	8	8	1	8x25+15x12
381	9	9	1	9x25+13x12
382	10	10	1	10x25+11x12
383	11	11	1	11x25+9x12
384	0	0	2	0x25+32x12 | 12x25+7x12
385	1	1	2	1x25+30x12 | 13x25+5x12
386	2	2	2	2x25+28x12 | 14x25+3x12
387	3	3	2	3x25+26x12 | 15x25+1x12
388	4	4	1	4x25+24x12
389	5	5	1	5x25+22x12
390	6	6	1	6x25+20x12
391	7	7	1	7x25+18x12
392	8	8	1	8x25+16x12
393	9	9	1	9x25+14x12
394	10	10	1	10x25+12x12
395	11	11	1	11x25+10x12
396	0	0	2	0x25+33x12 | 12x25+8x12
397	1	1	2	1x25+31x12 | 13x25+6x12
398	2	2	2	2x25+29x12 | 14x25+4x12
399	3	3	2	3x25+27x12 | 15x25+2x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
400	4	4	2	4x25+25x12 | 16x25+0x12
401	5	5	1	5x25+23x12
402	6	6	1	6x25+21x12
403	7	7	1	7x25+19x12
404	8	8	1	8x25+17x12
405	9	9	1	9x25+15x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
406	10	10	1	10x25+13x12
407	11	11	1	11x25+11x12
408	0	0	2	0x25+34x12	12x25+9x12
409	1	1	2	1x25+32x12	13x25+7x12
410	2	2	2	2x25+30x12	14x25+5x12
411	3	3	2	3x25+28x12	15x25+3x12
412	4	4	2	4x25+26x12	16x25+1x12
413	5	5	1	5x25+24x12
414	6	6	1	6x25+22x12
415	7	7	1	7x25+20x12
416	8	8	1	8x25+18x12
417	9	9	1	9x25+16x12
418	10	10	1	10x25+14x12
419	11	11	1	11x25+12x12
420	0	0	2	0x25+35x12	12x25+10x12
421	1	1	2	1x25+33x12	13x25+8x12
422	2	2	2	2x25+31x12	14x25+6x12
423	3	3	2	3x25+29x12	15x25+4x12
424	4	4	2	4x25+27x12	16x25+2x12
425	5	5	2	5x25+25x12	17x25+0x12
426	6	6	1	6x25+23x12
427	7	7	1	7x25+21x12
428	8	8	1	8x25+19x12
429	9	9	1	9x25+17x12
430	10	10	1	10x25+15x12
431	11	11	1	11x25+13x12
432	0	0	2	0x25+36x12	12x25+11x12
433	1	1	2	1x25+34x12	13x25+9x12
434	2	2	2	2x25+32x12	14x25+7x12
435	3	3	2	3x25+30x12	15x25+5x12
436	4	4	2	4x25+28x12	16x25+3x12
437	5	5	2	5x25+26x12	17x25+1x12

438	6	6	1	6x25+24x12
439	7	7	1	7x25+22x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
440	8	8	1	8x25+20x12
441	9	9	1	9x25+18x12
442	10	10	1	10x25+16x12
443	11	11	1	11x25+14x12
444	0	0	2	0x25+37x12 | 12x25+12x12
445	1	1	2	1x25+35x12 | 13x25+10x12
446	2	2	2	2x25+33x12 | 14x25+8x12
447	3	3	2	3x25+31x12 | 15x25+6x12
448	4	4	2	4x25+29x12 | 16x25+4x12
449	5	5	2	5x25+27x12 | 17x25+2x12
450	6	6	2	6x25+25x12 | 18x25+0x12
451	7	7	1	7x25+23x12
452	8	8	1	8x25+21x12
453	9	9	1	9x25+19x12
454	10	10	1	10x25+17x12
455	11	11	1	11x25+15x12
456	0	0	2	0x25+38x12 | 12x25+13x12
457	1	1	2	1x25+36x12 | 13x25+11x12
458	2	2	2	2x25+34x12 | 14x25+9x12
459	3	3	2	3x25+32x12 | 15x25+7x12
460	4	4	2	4x25+30x12 | 16x25+5x12
461	5	5	2	5x25+28x12 | 17x25+3x12
462	6	6	2	6x25+26x12 | 18x25+1x12
463	7	7	1	7x25+24x12
464	8	8	1	8x25+22x12
465	9	9	1	9x25+20x12
466	10	10	1	10x25+18x12
467	11	11	1	11x25+16x12
468	0	0	2	0x25+39x12 | 12x25+14x12
469	1	1	2	1x25+37x12 | 13x25+12x12
470	2	2	2	2x25+35x12 | 14x25+10x12
471	3	3	2	3x25+33x12 | 15x25+8x12
472	4	4	2	4x25+31x12 | 16x25+6x12
473	5	5	2	5x25+29x12 | 17x25+4x12
474	6	6	2	6x25+27x12 | 18x25+2x12
475	7	7	2	7x25+25x12 | 19x25+0x12
476	8	8	1	8x25+23x12

477	9	9	1	9x25+21x12
478	10	10	1	10x25+19x12
479	11	11	1	11x25+17x12
480	0	0	2	0x25+40x12 | 12x25+15x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
481	1	1	2	1x25+38x12 | 13x25+13x12
482	2	2	2	2x25+36x12 | 14x25+11x12
483	3	3	2	3x25+34x12 | 15x25+9x12
484	4	4	2	4x25+32x12 | 16x25+7x12
485	5	5	2	5x25+30x12 | 17x25+5x12
486	6	6	2	6x25+28x12 | 18x25+3x12
487	7	7	2	7x25+26x12 | 19x25+1x12
488	8	8	1	8x25+24x12
489	9	9	1	9x25+22x12
490	10	10	1	10x25+20x12
491	11	11	1	11x25+18x12
492	0	0	2	0x25+41x12 | 12x25+16x12
493	1	1	2	1x25+39x12 | 13x25+14x12
494	2	2	2	2x25+37x12 | 14x25+12x12
495	3	3	2	3x25+35x12 | 15x25+10x12
496	4	4	2	4x25+33x12 | 16x25+8x12
497	5	5	2	5x25+31x12 | 17x25+6x12
498	6	6	2	6x25+29x12 | 18x25+4x12
499	7	7	2	7x25+27x12 | 19x25+2x12
500	8	8	2	8x25+25x12 | 20x25+0x12
501	9	9	1	9x25+23x12
502	10	10	1	10x25+21x12
503	11	11	1	11x25+19x12
504	0	0	2	0x25+42x12 | 12x25+17x12
505	1	1	2	1x25+40x12 | 13x25+15x12
506	2	2	2	2x25+38x12 | 14x25+13x12
507	3	3	2	3x25+36x12 | 15x25+11x12
508	4	4	2	4x25+34x12 | 16x25+9x12
509	5	5	2	5x25+32x12 | 17x25+7x12
510	6	6	2	6x25+30x12 | 18x25+5x12
511	7	7	2	7x25+28x12 | 19x25+3x12
512	8	8	2	8x25+26x12 | 20x25+1x12
513	9	9	1	9x25+24x12
514	10	10	1	10x25+22x12
515	11	11	1	11x25+20x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
516	0	0	2	0x25+43x12 | 12x25+18x12
517	1	1	2	1x25+41x12 | 13x25+16x12
518	2	2	2	2x25+39x12 | 14x25+14x12
519	3	3	2	3x25+37x12 | 15x25+12x12
520	4	4	2	4x25+35x12 | 16x25+10x12
521	5	5	2	5x25+33x12 | 17x25+8x12
522	6	6	2	6x25+31x12 | 18x25+6x12
523	7	7	2	7x25+29x12 | 19x25+4x12
524	8	8	2	8x25+27x12 | 20x25+2x12
525	9	9	2	9x25+25x12 | 21x25+0x12
526	10	10	1	10x25+23x12
527	11	11	1	11x25+21x12
528	0	0	2	0x25+44x12 | 12x25+19x12
529	1	1	2	1x25+42x12 | 13x25+17x12
530	2	2	2	2x25+40x12 | 14x25+15x12
531	3	3	2	3x25+38x12 | 15x25+13x12
532	4	4	2	4x25+36x12 | 16x25+11x12
533	5	5	2	5x25+34x12 | 17x25+9x12
534	6	6	2	6x25+32x12 | 18x25+7x12
535	7	7	2	7x25+30x12 | 19x25+5x12
536	8	8	2	8x25+28x12 | 20x25+3x12
537	9	9	2	9x25+26x12 | 21x25+1x12
538	10	10	1	10x25+24x12
539	11	11	1	11x25+22x12
540	0	0	2	0x25+45x12 | 12x25+20x12
541	1	1	2	1x25+43x12 | 13x25+18x12
542	2	2	2	2x25+41x12 | 14x25+16x12
543	3	3	2	3x25+39x12 | 15x25+14x12
544	4	4	2	4x25+37x12 | 16x25+12x12
545	5	5	2	5x25+35x12 | 17x25+10x12
546	6	6	2	6x25+33x12 | 18x25+8x12
547	7	7	2	7x25+31x12 | 19x25+6x12

548	8	8	2	8x25+29x12 | 20x25+4x12
549	9	9	2	9x25+27x12 | 21x25+2x12
550	10	10	2	10x25+25x12 | 22x25+0x12
551	11	11	1	11x25+23x12
552	0	0	2	0x25+46x12 | 12x25+21x12
553	1	1	2	1x25+44x12 | 13x25+19x12
554	2	2	2	2x25+42x12 | 14x25+17x12
555	3	3	2	3x25+40x12 | 15x25+15x12
556	4	4	2	4x25+38x12 | 16x25+13x12
557	5	5	2	5x25+36x12 | 17x25+11x12
558	6	6	2	6x25+34x12 | 18x25+9x12
559	7	7	2	7x25+32x12 | 19x25+7x12
560	8	8	2	8x25+30x12 | 20x25+5x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
561	9	9	2	9x25+28x12 | 21x25+3x12
562	10	10	2	10x25+26x12 | 22x25+1x12
563	11	11	1	11x25+24x12
564	0	0	2	0x25+47x12 | 12x25+22x12
565	1	1	2	1x25+45x12 | 13x25+20x12
566	2	2	2	2x25+43x12 | 14x25+18x12
567	3	3	2	3x25+41x12 | 15x25+16x12
568	4	4	2	4x25+39x12 | 16x25+14x12
569	5	5	2	5x25+37x12 | 17x25+12x12
570	6	6	2	6x25+35x12 | 18x25+10x12
571	7	7	2	7x25+33x12 | 19x25+8x12
572	8	8	2	8x25+31x12 | 20x25+6x12
573	9	9	2	9x25+29x12 | 21x25+4x12
574	10	10	2	10x25+27x12 | 22x25+2x12
575	11	11	2	11x25+25x12 | 23x25+0x12
576	0	0	2	0x25+48x12 | 12x25+23x12
577	1	1	2	1x25+46x12 | 13x25+21x12
578	2	2	2	2x25+44x12 | 14x25+19x12
579	3	3	2	3x25+42x12 | 15x25+17x12
580	4	4	2	4x25+40x12 | 16x25+15x12
581	5	5	2	5x25+38x12 | 17x25+13x12
582	6	6	2	6x25+36x12 | 18x25+11x12
583	7	7	2	7x25+34x12 | 19x25+9x12
584	8	8	2	8x25+32x12 | 20x25+7x12

585	9	9	2	9x25+30x12 | 21x25+5x12
586	10	10	2	10x25+28x12 | 22x25+3x12
587	11	11	2	11x25+26x12 | 23x25+1x12
588	0	0	2	0x25+49x12 | 12x25+24x12
589	1	1	2	1x25+47x12 | 13x25+22x12
590	2	2	2	2x25+45x12 | 14x25+20x12
591	3	3	2	3x25+43x12 | 15x25+18x12
592	4	4	2	4x25+41x12 | 16x25+16x12
593	5	5	2	5x25+39x12 | 17x25+14x12
594	6	6	2	6x25+37x12 | 18x25+12x12
595	7	7	2	7x25+35x12 | 19x25+10x12
596	8	8	2	8x25+33x12 | 20x25+8x12
597	9	9	2	9x25+31x12 | 21x25+6x12
598	10	10	2	10x25+29x12 | 22x25+4x12
599	11	11	2	11x25+27x12 | 23x25+2x12
600	0	0	3	0x25+50x12 | 12x25+25x12 | 24x25+0x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
601	1	1	2	1x25+48x12 | 13x25+23x12
602	2	2	2	2x25+46x12 | 14x25+21x12
603	3	3	2	3x25+44x12 | 15x25+19x12
604	4	4	2	4x25+42x12 | 16x25+17x12
605	5	5	2	5x25+40x12 | 17x25+15x12
606	6	6	2	6x25+38x12 | 18x25+13x12
607	7	7	2	7x25+36x12 | 19x25+11x12
608	8	8	2	8x25+34x12 | 20x25+9x12
609	9	9	2	9x25+32x12 | 21x25+7x12
610	10	10	2	10x25+30x12 | 22x25+5x12
611	11	11	2	11x25+28x12 | 23x25+3x12
612	0	0	3	0x25+51x12 | 12x25+26x12 | 24x25+1x12
613	1	1	2	1x25+49x12 | 13x25+24x12
614	2	2	2	2x25+47x12 | 14x25+22x12
615	3	3	2	3x25+45x12 | 15x25+20x12
616	4	4	2	4x25+43x12 | 16x25+18x12
617	5	5	2	5x25+41x12 | 17x25+16x12
618	6	6	2	6x25+39x12 | 18x25+14x12
619	7	7	2	7x25+37x12 | 19x25+12x12
620	8	8	2	8x25+35x12 | 20x25+10x12
621	9	9	2	9x25+33x12 | 21x25+8x12
622	10	10	2	10x25+31x12 | 22x25+6x12
623	11	11	2	11x25+29x12 | 23x25+4x12

624	0	0	3	0x25+52x12 | 12x25+27x12 | 24x25+2x12
625	1	1	3	1x25+50x12 | 13x25+25x12 | 25x25+0x12
626	2	2	2	2x25+48x12 | 14x25+23x12
627	3	3	2	3x25+46x12 | 15x25+21x12
628	4	4	2	4x25+44x12 | 16x25+19x12
629	5	5	2	5x25+42x12 | 17x25+17x12
630	6	6	2	6x25+40x12 | 18x25+15x12
631	7	7	2	7x25+38x12 | 19x25+13x12
632	8	8	2	8x25+36x12 | 20x25+11x12
633	9	9	2	9x25+34x12 | 21x25+9x12
634	10	10	2	10x25+32x12 | 22x25+7x12
635	11	11	2	11x25+30x12 | 23x25+5x12
636	0	0	3	0x25+53x12 | 12x25+28x12 | 24x25+3x12
637	1	1	3	1x25+51x12 | 13x25+26x12 | 25x25+1x12
638	2	2	2	2x25+49x12 | 14x25+24x12
639	3	3	2	3x25+47x12 | 15x25+22x12
640	4	4	2	4x25+45x12 | 16x25+20x12
641	5	5	2	5x25+43x12 | 17x25+18x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
642	6	6	2	6x25+41x12 | 18x25+16x12
643	7	7	2	7x25+39x12 | 19x25+14x12
644	8	8	2	8x25+37x12 | 20x25+12x12
645	9	9	2	9x25+35x12 | 21x25+10x12
646	10	10	2	10x25+33x12 | 22x25+8x12
647	11	11	2	11x25+31x12 | 23x25+6x12
648	0	0	3	0x25+54x12 | 12x25+29x12 | 24x25+4x12
649	1	1	3	1x25+52x12 | 13x25+27x12 | 25x25+2x12
650	2	2	3	2x25+50x12 | 14x25+25x12 | 26x25+0x12
651	3	3	2	3x25+48x12 | 15x25+23x12
652	4	4	2	4x25+46x12 | 16x25+21x12
653	5	5	2	5x25+44x12 | 17x25+19x12
654	6	6	2	6x25+42x12 | 18x25+17x12
655	7	7	2	7x25+40x12 | 19x25+15x12
656	8	8	2	8x25+38x12 | 20x25+13x12
657	9	9	2	9x25+36x12 | 21x25+11x12
658	10	10	2	10x25+34x12 | 22x25+9x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
659	11	11	2	11x25+32x12 | 23x25+7x12
660	0	0	3	0x25+55x12 | 12x25+30x12 | 24x25+5x12
661	1	1	3	1x25+53x12 | 13x25+28x12 | 25x25+3x12
662	2	2	3	2x25+51x12 | 14x25+26x12 | 26x25+1x12
663	3	3	2	3x25+49x12 | 15x25+24x12
664	4	4	2	4x25+47x12 | 16x25+22x12
665	5	5	2	5x25+45x12 | 17x25+20x12
666	6	6	2	6x25+43x12 | 18x25+18x12
667	7	7	2	7x25+41x12 | 19x25+16x12
668	8	8	2	8x25+39x12 | 20x25+14x12
669	9	9	2	9x25+37x12 | 21x25+12x12
670	10	10	2	10x25+35x12 | 22x25+10x12
671	11	11	2	11x25+33x12 | 23x25+8x12
672	0	0	3	0x25+56x12 | 12x25+31x12 | 24x25+6x12
673	1	1	3	1x25+54x12 | 13x25+29x12 | 25x25+4x12
674	2	2	3	2x25+52x12 | 14x25+27x12 | 26x25+2x12
675	3	3	3	3x25+50x12 | 15x25+25x12 | 27x25+0x12
676	4	4	2	4x25+48x12 | 16x25+23x12
677	5	5	2	5x25+46x12 | 17x25+21x12
678	6	6	2	6x25+44x12 | 18x25+19x12
679	7	7	2	7x25+42x12 | 19x25+17x12
680	8	8	2	8x25+40x12 | 20x25+15x12
681	9	9	2	9x25+38x12 | 21x25+13x12
682	10	10	2	10x25+36x12 | 22x25+11x12
683	11	11	2	11x25+34x12 | 23x25+9x12
684	0	0	3	0x25+57x12 | 12x25+32x12 | 24x25+7x12
685	1	1	3	1x25+55x12 | 13x25+30x12 | 25x25+5x12
686	2	2	3	2x25+53x12 | 14x25+28x12 | 26x25+3x12
687	3	3	3	3x25+51x12 | 15x25+26x12 | 27x25+1x12
688	4	4	2	4x25+49x12 | 16x25+24x12
689	5	5	2	5x25+47x12 | 17x25+22x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
690	6	6	2	6x25+45x12 | 18x25+20x12
691	7	7	2	7x25+43x12 | 19x25+18x12
692	8	8	2	8x25+41x12 | 20x25+16x12
693	9	9	2	9x25+39x12 | 21x25+14x12
694	10	10	2	10x25+37x12 | 22x25+12x12
695	11	11	2	11x25+35x12 | 23x25+10x12
696	0	0	3	0x25+58x12 | 12x25+33x12 | 24x25+8x12
697	1	1	3	1x25+56x12 | 13x25+31x12 | 25x25+6x12
698	2	2	3	2x25+54x12 | 14x25+29x12 | 26x25+4x12
699	3	3	3	3x25+52x12 | 15x25+27x12 | 27x25+2x12
700	4	4	3	4x25+50x12 | 16x25+25x12 | 28x25+0x12
701	5	5	2	5x25+48x12 | 17x25+23x12
702	6	6	2	6x25+46x12 | 18x25+21x12
703	7	7	2	7x25+44x12 | 19x25+19x12
704	8	8	2	8x25+42x12 | 20x25+17x12
705	9	9	2	9x25+40x12 | 21x25+15x12
706	10	10	2	10x25+38x12 | 22x25+13x12
707	11	11	2	11x25+36x12 | 23x25+11x12
708	0	0	3	0x25+59x12 | 12x25+34x12 | 24x25+9x12
709	1	1	3	1x25+57x12 | 13x25+32x12 | 25x25+7x12
710	2	2	3	2x25+55x12 | 14x25+30x12 | 26x25+5x12
711	3	3	3	3x25+53x12 | 15x25+28x12 | 27x25+3x12
712	4	4	3	4x25+51x12 | 16x25+26x12 | 28x25+1x12
713	5	5	2	5x25+49x12 | 17x25+24x12
714	6	6	2	6x25+47x12 | 18x25+22x12
715	7	7	2	7x25+45x12 | 19x25+20x12
716	8	8	2	8x25+43x12 | 20x25+18x12
717	9	9	2	9x25+41x12 | 21x25+16x12
718	10	10	2	10x25+39x12 | 22x25+14x12
719	11	11	2	11x25+37x12 | 23x25+12x12
720	0	0	3	0x25+60x12 | 12x25+35x12 | 24x25+10x12
721	1	1	3	1x25+58x12 | 13x25+33x12 | 25x25+8x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
722	2	2	3	2x25+56x12 | 14x25+31x12 | 26x25+6x12
723	3	3	3	3x25+54x12 | 15x25+29x12 | 27x25+4x12
724	4	4	3	4x25+52x12 | 16x25+27x12 | 28x25+2x12
725	5	5	3	5x25+50x12 | 17x25+25x12 | 29x25+0x12
726	6	6	2	6x25+48x12 | 18x25+23x12
727	7	7	2	7x25+46x12 | 19x25+21x12
728	8	8	2	8x25+44x12 | 20x25+19x12
729	9	9	2	9x25+42x12 | 21x25+17x12
730	10	10	2	10x25+40x12 | 22x25+15x12
731	11	11	2	11x25+38x12 | 23x25+13x12
732	0	0	3	0x25+61x12 | 12x25+36x12 | 24x25+11x12
733	1	1	3	1x25+59x12 | 13x25+34x12 | 25x25+9x12
734	2	2	3	2x25+57x12 | 14x25+32x12 | 26x25+7x12
735	3	3	3	3x25+55x12 | 15x25+30x12 | 27x25+5x12
736	4	4	3	4x25+53x12 | 16x25+28x12 | 28x25+3x12
737	5	5	3	5x25+51x12 | 17x25+26x12 | 29x25+1x12
738	6	6	2	6x25+49x12 | 18x25+24x12
739	7	7	2	7x25+47x12 | 19x25+22x12
740	8	8	2	8x25+45x12 | 20x25+20x12
741	9	9	2	9x25+43x12 | 21x25+18x12
742	10	10	2	10x25+41x12 | 22x25+16x12
743	11	11	2	11x25+39x12 | 23x25+14x12
744	0	0	3	0x25+62x12 | 12x25+37x12 | 24x25+12x12
745	1	1	3	1x25+60x12 | 13x25+35x12 | 25x25+10x12
746	2	2	3	2x25+58x12 | 14x25+33x12 | 26x25+8x12
747	3	3	3	3x25+56x12 | 15x25+31x12 | 27x25+6x12
748	4	4	3	4x25+54x12 | 16x25+29x12 | 28x25+4x12
749	5	5	3	5x25+52x12 | 17x25+27x12 | 29x25+2x12
750	6	6	3	6x25+50x12 | 18x25+25x12 | 30x25+0x12
751	7	7	2	7x25+48x12 | 19x25+23x12
752	8	8	2	8x25+46x12 | 20x25+21x12
753	9	9	2	9x25+44x12 | 21x25+19x12
754	10	10	2	10x25+42x12 | 22x25+17x12
755	11	11	2	11x25+40x12 | 23x25+15x12
756	0	0	3	0x25+63x12 | 12x25+38x12 | 24x25+13x12
757	1	1	3	1x25+61x12 | 13x25+36x12 | 25x25+11x12
758	2	2	3	2x25+59x12 | 14x25+34x12 | 26x25+9x12
759	3	3	3	3x25+57x12 | 15x25+32x12 | 27x25+7x12
760	4	4	3	4x25+55x12 | 16x25+30x12 | 28x25+5x12
761	5	5	3	5x25+53x12 | 17x25+28x12 | 29x25+3x12

762	6	6	3	6x25+51x12 | 18x25+26x12 | 30x25+1x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
763	7	7	2	7x25+49x12 | 19x25+24x12
764	8	8	2	8x25+47x12 | 20x25+22x12
765	9	9	2	9x25+45x12 | 21x25+20x12
766	10	10	2	10x25+43x12 | 22x25+18x12
767	11	11	2	11x25+41x12 | 23x25+16x12
768	0	0	3	0x25+64x12 | 12x25+39x12 | 24x25+14x12
769	1	1	3	1x25+62x12 | 13x25+37x12 | 25x25+12x12
770	2	2	3	2x25+60x12 | 14x25+35x12 | 26x25+10x12
771	3	3	3	3x25+58x12 | 15x25+33x12 | 27x25+8x12
772	4	4	3	4x25+56x12 | 16x25+31x12 | 28x25+6x12
773	5	5	3	5x25+54x12 | 17x25+29x12 | 29x25+4x12
774	6	6	3	6x25+52x12 | 18x25+27x12 | 30x25+2x12
775	7	7	3	7x25+50x12 | 19x25+25x12 | 31x25+0x12
776	8	8	2	8x25+48x12 | 20x25+23x12
777	9	9	2	9x25+46x12 | 21x25+21x12
778	10	10	2	10x25+44x12 | 22x25+19x12
779	11	11	2	11x25+42x12 | 23x25+17x12
780	0	0	3	0x25+65x12 | 12x25+40x12 | 24x25+15x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
781	1	1	3	1x25+63x12 | 13x25+38x12 | 25x25+13x12
782	2	2	3	2x25+61x12 | 14x25+36x12 | 26x25+11x12
783	3	3	3	3x25+59x12 | 15x25+34x12 | 27x25+9x12
784	4	4	3	4x25+57x12 | 16x25+32x12 | 28x25+7x12
785	5	5	3	5x25+55x12 | 17x25+30x12 | 29x25+5x12
786	6	6	3	6x25+53x12 | 18x25+28x12 | 30x25+3x12
787	7	7	3	7x25+51x12 | 19x25+26x12 | 31x25+1x12
788	8	8	2	8x25+49x12 | 20x25+24x12
789	9	9	2	9x25+47x12 | 21x25+22x12
790	10	10	2	10x25+45x12 | 22x25+20x12
791	11	11	2	11x25+43x12 | 23x25+18x12
792	0	0	3	0x25+66x12 | 12x25+41x12 | 24x25+16x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
793	1	1	3	1x25+64x12 | 13x25+39x12 | 25x25+14x12
794	2	2	3	2x25+62x12 | 14x25+37x12 | 26x25+12x12
795	3	3	3	3x25+60x12 | 15x25+35x12 | 27x25+10x12
796	4	4	3	4x25+58x12 | 16x25+33x12 | 28x25+8x12
797	5	5	3	5x25+56x12 | 17x25+31x12 | 29x25+6x12
798	6	6	3	6x25+54x12 | 18x25+29x12 | 30x25+4x12
799	7	7	3	7x25+52x12 | 19x25+27x12 | 31x25+2x12
800	8	8	3	8x25+50x12 | 20x25+25x12 | 32x25+0x12
801	9	9	2	9x25+48x12 | 21x25+23x12
802	10	10	2	10x25+46x12 | 22x25+21x12
803	11	11	2	11x25+44x12 | 23x25+19x12
804	0	0	3	0x25+67x12 | 12x25+42x12 | 24x25+17x12
805	1	1	3	1x25+65x12 | 13x25+40x12 | 25x25+15x12
806	2	2	3	2x25+63x12 | 14x25+38x12 | 26x25+13x12
807	3	3	3	3x25+61x12 | 15x25+36x12 | 27x25+11x12
808	4	4	3	4x25+59x12 | 16x25+34x12 | 28x25+9x12
809	5	5	3	5x25+57x12 | 17x25+32x12 | 29x25+7x12
810	6	6	3	6x25+55x12 | 18x25+30x12 | 30x25+5x12
811	7	7	3	7x25+53x12 | 19x25+28x12 | 31x25+3x12
812	8	8	3	8x25+51x12 | 20x25+26x12 | 32x25+1x12
813	9	9	2	9x25+49x12 | 21x25+24x12
814	10	10	2	10x25+47x12 | 22x25+22x12
815	11	11	2	11x25+45x12 | 23x25+20x12
816	0	0	3	0x25+68x12 | 12x25+43x12 | 24x25+18x12
817	1	1	3	1x25+66x12 | 13x25+41x12 | 25x25+16x12
818	2	2	3	2x25+64x12 | 14x25+39x12 | 26x25+14x12
819	3	3	3	3x25+62x12 | 15x25+37x12 | 27x25+12x12
820	4	4	3	4x25+60x12 | 16x25+35x12 | 28x25+10x12
821	5	5	3	5x25+58x12 | 17x25+33x12 | 29x25+8x12
822	6	6	3	6x25+56x12 | 18x25+31x12 | 30x25+6x12
823	7	7	3	7x25+54x12 | 19x25+29x12 | 31x25+4x12
824	8	8	3	8x25+52x12 | 20x25+27x12 | 32x25+2x12

825	9	9	3	9x25+50x12 | 21x25+25x12 | 33x25+0x12
826	10	10	2	10x25+48x12 | 22x25+23x12
827	11	11	2	11x25+46x12 | 23x25+21x12
828	0	0	3	0x25+69x12 | 12x25+44x12 | 24x25+19x12
829	1	1	3	1x25+67x12 | 13x25+42x12 | 25x25+17x12
830	2	2	3	2x25+65x12 | 14x25+40x12 | 26x25+15x12
831	3	3	3	3x25+63x12 | 15x25+38x12 | 27x25+13x12
832	4	4	3	4x25+61x12 | 16x25+36x12 | 28x25+11x12
833	5	5	3	5x25+59x12 | 17x25+34x12 | 29x25+9x12
834	6	6	3	6x25+57x12 | 18x25+32x12 | 30x25+7x12
835	7	7	3	7x25+55x12 | 19x25+30x12 | 31x25+5x12
836	8	8	3	8x25+53x12 | 20x25+28x12 | 32x25+3x12
837	9	9	3	9x25+51x12 | 21x25+26x12 | 33x25+1x12
838	10	10	2	10x25+49x12 | 22x25+24x12
839	11	11	2	11x25+47x12 | 23x25+22x12
840	0	0	3	0x25+70x12 | 12x25+45x12 | 24x25+20x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
841	1	1	3	1x25+68x12 | 13x25+43x12 | 25x25+18x12
842	2	2	3	2x25+66x12 | 14x25+41x12 | 26x25+16x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
843	3	3	3	3x25+64x12 | 15x25+39x12 | 27x25+14x12
844	4	4	3	4x25+62x12 | 16x25+37x12 | 28x25+12x12
845	5	5	3	5x25+60x12 | 17x25+35x12 | 29x25+10x12
846	6	6	3	6x25+58x12 | 18x25+33x12 | 30x25+8x12
847	7	7	3	7x25+56x12 | 19x25+31x12 | 31x25+6x12
848	8	8	3	8x25+54x12 | 20x25+29x12 | 32x25+4x12
849	9	9	3	9x25+52x12 | 21x25+27x12 | 33x25+2x12
850	10	10	3	10x25+50x12 | 22x25+25x12 | 34x25+0x12
851	11	11	2	11x25+48x12 | 23x25+23x12
852	0	0	3	0x25+71x12 | 12x25+46x12 | 24x25+21x12
853	1	1	3	1x25+69x12 | 13x25+44x12 | 25x25+19x12
854	2	2	3	2x25+67x12 | 14x25+42x12 | 26x25+17x12
855	3	3	3	3x25+65x12 | 15x25+40x12 | 27x25+15x12
856	4	4	3	4x25+63x12 | 16x25+38x12 | 28x25+13x12
857	5	5	3	5x25+61x12 | 17x25+36x12 | 29x25+11x12
858	6	6	3	6x25+59x12 | 18x25+34x12 | 30x25+9x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
859	7	7	3	7x25+57x12 | 19x25+32x12 | 31x25+7x12
860	8	8	3	8x25+55x12 | 20x25+30x12 | 32x25+5x12
861	9	9	3	9x25+53x12 | 21x25+28x12 | 33x25+3x12
862	10	10	3	10x25+51x12 | 22x25+26x12 | 34x25+1x12
863	11	11	2	11x25+49x12 | 23x25+24x12
864	0	0	3	0x25+72x12 | 12x25+47x12 | 24x25+22x12
865	1	1	3	1x25+70x12 | 13x25+45x12 | 25x25+20x12
866	2	2	3	2x25+68x12 | 14x25+43x12 | 26x25+18x12
867	3	3	3	3x25+66x12 | 15x25+41x12 | 27x25+16x12
868	4	4	3	4x25+64x12 | 16x25+39x12 | 28x25+14x12
869	5	5	3	5x25+62x12 | 17x25+37x12 | 29x25+12x12
870	6	6	3	6x25+60x12 | 18x25+35x12 | 30x25+10x12
871	7	7	3	7x25+58x12 | 19x25+33x12 | 31x25+8x12
872	8	8	3	8x25+56x12 | 20x25+31x12 | 32x25+6x12
873	9	9	3	9x25+54x12 | 21x25+29x12 | 33x25+4x12
874	10	10	3	10x25+52x12 | 22x25+27x12 | 34x25+2x12
875	11	11	3	11x25+50x12 | 23x25+25x12 | 35x25+0x12
876	0	0	3	0x25+73x12 | 12x25+48x12 | 24x25+23x12
877	1	1	3	1x25+71x12 | 13x25+46x12 | 25x25+21x12
878	2	2	3	2x25+69x12 | 14x25+44x12 | 26x25+19x12
879	3	3	3	3x25+67x12 | 15x25+42x12 | 27x25+17x12
880	4	4	3	4x25+65x12 | 16x25+40x12 | 28x25+15x12
881	5	5	3	5x25+63x12 | 17x25+38x12 | 29x25+13x12
882	6	6	3	6x25+61x12 | 18x25+36x12 | 30x25+11x12
883	7	7	3	7x25+59x12 | 19x25+34x12 | 31x25+9x12
884	8	8	3	8x25+57x12 | 20x25+32x12 | 32x25+7x12
885	9	9	3	9x25+55x12 | 21x25+30x12 | 33x25+5x12
886	10	10	3	10x25+53x12 | 22x25+28x12 | 34x25+3x12
887	11	11	3	11x25+51x12 | 23x25+26x12 | 35x25+1x12
888	0	0	3	0x25+74x12 | 12x25+49x12 | 24x25+24x12
889	1	1	3	1x25+72x12 | 13x25+47x12 | 25x25+22x12
890	2	2	3	2x25+70x12 | 14x25+45x12 | 26x25+20x12
891	3	3	3	3x25+68x12 | 15x25+43x12 | 27x25+18x12
892	4	4	3	4x25+66x12 | 16x25+41x12 | 28x25+16x12
893	5	5	3	5x25+64x12 | 17x25+39x12 | 29x25+14x12
894	6	6	3	6x25+62x12 | 18x25+37x12 | 30x25+12x12
895	7	7	3	7x25+60x12 | 19x25+35x12 | 31x25+10x12

896	8	8	3	8x25+58x12 | 20x25+33x12 | 32x25+8x12
897	9	9	3	9x25+56x12 | 21x25+31x12 | 33x25+6x12
898	10	10	3	10x25+54x12 | 22x25+29x12 | 34x25+4x12
899	11	11	3	11x25+52x12 | 23x25+27x12 | 35x25+2x12
900	0	0	4	0x25+75x12 | 12x25+50x12 | 24x25+25x12 ... (+1)

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
901	1	1	3	1x25+73x12 | 13x25+48x12 | 25x25+23x12
902	2	2	3	2x25+71x12 | 14x25+46x12 | 26x25+21x12
903	3	3	3	3x25+69x12 | 15x25+44x12 | 27x25+19x12
904	4	4	3	4x25+67x12 | 16x25+42x12 | 28x25+17x12
905	5	5	3	5x25+65x12 | 17x25+40x12 | 29x25+15x12
906	6	6	3	6x25+63x12 | 18x25+38x12 | 30x25+13x12
907	7	7	3	7x25+61x12 | 19x25+36x12 | 31x25+11x12
908	8	8	3	8x25+59x12 | 20x25+34x12 | 32x25+9x12
909	9	9	3	9x25+57x12 | 21x25+32x12 | 33x25+7x12
910	10	10	3	10x25+55x12 | 22x25+30x12 | 34x25+5x12
911	11	11	3	11x25+53x12 | 23x25+28x12 | 35x25+3x12
912	0	0	4	0x25+76x12 | 12x25+51x12 | 24x25+26x12 ... (+1)
913	1	1	3	1x25+74x12 | 13x25+49x12 | 25x25+24x12
914	2	2	3	2x25+72x12 | 14x25+47x12 | 26x25+22x12
915	3	3	3	3x25+70x12 | 15x25+45x12 | 27x25+20x12
916	4	4	3	4x25+68x12 | 16x25+43x12 | 28x25+18x12
917	5	5	3	5x25+66x12 | 17x25+41x12 | 29x25+16x12
918	6	6	3	6x25+64x12 | 18x25+39x12 | 30x25+14x12
919	7	7	3	7x25+62x12 | 19x25+37x12 | 31x25+12x12
920	8	8	3	8x25+60x12 | 20x25+35x12 | 32x25+10x12
921	9	9	3	9x25+58x12 | 21x25+33x12 | 33x25+8x12
922	10	10	3	10x25+56x12 | 22x25+31x12 | 34x25+6x12
923	11	11	3	11x25+54x12 | 23x25+29x12 | 35x25+4x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
924	0	0	4	0x25+77x12 | 12x25+52x12 | 24x25+27x12 ... (+1)
925	1	1	4	1x25+75x12 | 13x25+50x12 | 25x25+25x12 ... (+1)
926	2	2	3	2x25+73x12 | 14x25+48x12 | 26x25+23x12

927	3	3	3	3x25+71x12 | 15x25+46x12 | 27x25+21x12
928	4	4	3	4x25+69x12 | 16x25+44x12 | 28x25+19x12
929	5	5	3	5x25+67x12 | 17x25+42x12 | 29x25+17x12
930	6	6	3	6x25+65x12 | 18x25+40x12 | 30x25+15x12
931	7	7	3	7x25+63x12 | 19x25+38x12 | 31x25+13x12
932	8	8	3	8x25+61x12 | 20x25+36x12 | 32x25+11x12
933	9	9	3	9x25+59x12 | 21x25+34x12 | 33x25+9x12
934	10	10	3	10x25+57x12 | 22x25+32x12 | 34x25+7x12
935	11	11	3	11x25+55x12 | 23x25+30x12 | 35x25+5x12
936	0	0	4	0x25+78x12 | 12x25+53x12 | 24x25+28x12 ... (+1)
937	1	1	4	1x25+76x12 | 13x25+51x12 | 25x25+26x12 ... (+1)
938	2	2	3	2x25+74x12 | 14x25+49x12 | 26x25+24x12
939	3	3	3	3x25+72x12 | 15x25+47x12 | 27x25+22x12
940	4	4	3	4x25+70x12 | 16x25+45x12 | 28x25+20x12
941	5	5	3	5x25+68x12 | 17x25+43x12 | 29x25+18x12
942	6	6	3	6x25+66x12 | 18x25+41x12 | 30x25+16x12
943	7	7	3	7x25+64x12 | 19x25+39x12 | 31x25+14x12
944	8	8	3	8x25+62x12 | 20x25+37x12 | 32x25+12x12
945	9	9	3	9x25+60x12 | 21x25+35x12 | 33x25+10x12
946	10	10	3	10x25+58x12 | 22x25+33x12 | 34x25+8x12
947	11	11	3	11x25+56x12 | 23x25+31x12 | 35x25+6x12
948	0	0	4	0x25+79x12 | 12x25+54x12 | 24x25+29x12 ... (+1)
949	1	1	4	1x25+77x12 | 13x25+52x12 | 25x25+27x12 ... (+1)
950	2	2	4	2x25+75x12 | 14x25+50x12 | 26x25+25x12 ... (+1)
951	3	3	3	3x25+73x12 | 15x25+48x12 | 27x25+23x12
952	4	4	3	4x25+71x12 | 16x25+46x12 | 28x25+21x12
953	5	5	3	5x25+69x12 | 17x25+44x12 | 29x25+19x12
954	6	6	3	6x25+67x12 | 18x25+42x12 | 30x25+17x12
955	7	7	3	7x25+65x12 | 19x25+40x12 | 31x25+15x12
956	8	8	3	8x25+63x12 | 20x25+38x12 | 32x25+13x12
957	9	9	3	9x25+61x12 | 21x25+36x12 | 33x25+11x12
958	10	10	3	10x25+59x12 | 22x25+34x12 | 34x25+9x12
959	11	11	3	11x25+57x12 | 23x25+32x12 | 35x25+7x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
960	0	0	4	0x25+80x12 | 12x25+55x12 | 24x25+30x12 ... (+1)

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
961	1	1	4	1x25+78x12 | 13x25+53x12 | 25x25+28x12 ... (+1)
962	2	2	4	2x25+76x12 | 14x25+51x12 | 26x25+26x12 ... (+1)
963	3	3	3	3x25+74x12 | 15x25+49x12 | 27x25+24x12

N	r=N mod 12	A0	R(N)	Minimal + next representation(s)
964	4	4	3	4x25+72x12 | 16x25+47x12 | 28x25+22x12
965	5	5	3	5x25+70x12 | 17x25+45x12 | 29x25+20x12
966	6	6	3	6x25+68x12 | 18x25+43x12 | 30x25+18x12
967	7	7	3	7x25+66x12 | 19x25+41x12 | 31x25+16x12
968	8	8	3	8x25+64x12 | 20x25+39x12 | 32x25+14x12
969	9	9	3	9x25+62x12 | 21x25+37x12 | 33x25+12x12
970	10	10	3	10x25+60x12 | 22x25+35x12 | 34x25+10x12
971	11	11	3	11x25+58x12 | 23x25+33x12 | 35x25+8x12
972	0	0	4	0x25+81x12 | 12x25+56x12 | 24x25+31x12 ... (+1)
973	1	1	4	1x25+79x12 | 13x25+54x12 | 25x25+29x12 ... (+1)
974	2	2	4	2x25+77x12 | 14x25+52x12 | 26x25+27x12 ... (+1)
975	3	3	4	3x25+75x12 | 15x25+50x12 | 27x25+25x12 ... (+1)
976	4	4	3	4x25+73x12 | 16x25+48x12 | 28x25+23x12
977	5	5	3	5x25+71x12 | 17x25+46x12 | 29x25+21x12
978	6	6	3	6x25+69x12 | 18x25+44x12 | 30x25+19x12
979	7	7	3	7x25+67x12 | 19x25+42x12 | 31x25+17x12
980	8	8	3	8x25+65x12 | 20x25+40x12 | 32x25+15x12
981	9	9	3	9x25+63x12 | 21x25+38x12 | 33x25+13x12
982	10	10	3	10x25+61x12 | 22x25+36x12 | 34x25+11x12
983	11	11	3	11x25+59x12 | 23x25+34x12 | 35x25+9x12
984	0	0	4	0x25+82x12 | 12x25+57x12 | 24x25+32x12 ... (+1)
985	1	1	4	1x25+80x12 | 13x25+55x12 | 25x25+30x12 ... (+1)
986	2	2	4	2x25+78x12 | 14x25+53x12 | 26x25+28x12 ... (+1)
987	3	3	4	3x25+76x12 | 15x25+51x12 | 27x25+26x12 ... (+1)
988	4	4	3	4x25+74x12 | 16x25+49x12 | 28x25+24x12
989	5	5	3	5x25+72x12 | 17x25+47x12 | 29x25+22x12
990	6	6	3	6x25+70x12 | 18x25+45x12 | 30x25+20x12
991	7	7	3	7x25+68x12 | 19x25+43x12 | 31x25+18x12
992	8	8	3	8x25+66x12 | 20x25+41x12 | 32x25+16x12
993	9	9	3	9x25+64x12 | 21x25+39x12 | 33x25+14x12
994	10	10	3	10x25+62x12 | 22x25+37x12 | 34x25+12x12
995	11	11	3	11x25+60x12 | 23x25+35x12 | 35x25+10x12
996	0	0	4	0x25+83x12 | 12x25+58x12 | 24x25+33x12 ... (+1)

997	1	1	4	1x25+81x12 | 13x25+56x12 | 25x25+31x12 ... (+1)
998	2	2	4	2x25+79x12 | 14x25+54x12 | 26x25+29x12 ... (+1)
999	3	3	4	3x25+77x12 | 15x25+52x12 | 27x25+27x12 ... (+1)
1000	4	4	4	4x25+75x12 | 16x25+50x12 | 28x25+25x12 ... (+1)

Epilogue -- The Structure Closes

De volledige theorie rust op een structurele voorwaarde: p = 1 (mod q). Voor het (25,12)-systeem levert dit dezelfde geometrie als het (19,9)-systeem, op een andere schaal:

1. **B verdwijnt mod q:** 25A + 12B = N (mod 12) wordt A = N (mod 12), such that A0 = N mod 12 direct bepaalbaar is.
2. **Ladder-structuur:** All representations vormen een rekenkundige ladder met stap 12 in A en stap -25 in B.
3. **Periode = pxq = 300:** Elke 300 eenheden stijgt het minimum aantal representations met 1.

Familie-grenzen G(k) = 300k - 288.

4. **r=11 is de laatste:** Precies zoals dr=8 in het (19,9)-systeem, is r=11 de residue class die het langst wacht op een representation en structureel een stap achterloopt.

The Fundamental Equation

$$N = 25A + 12B$$

$$A0 = N \bmod 12$$

$$R(N) = \text{floor}((\text{floor}(N/25) - A0) / 12) + 1$$

A0 = N mod 12 | Bilal El Issaoui | Amsterdam | 2026